What factors from the NFL combine and draft process best predict future success?

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Project Objectives and Questions

Objective(s):

- Develop a predictive model to estimate the likelihood of NFL success for 2025 and future draft picks using an optimal combination of NFL Combine metrics. Create models using different techniques and compare to see which is the best.

Question(s):

- Which variables best predict NFL success? Are NFL Combine metrics actually important? Which players are undervalued in the draft? Which combine drills are overvalued?

Techniques Used

- Linear Regression (OLS)
- LASSO
- Ridge
- Random Forest
  - Models were compared using 10-fold cross validation (5 repeats)
  - RMSE and R^2 will be used to compare model performance

Data Sources

NFL Combine Statistics (Kaggle)

- Player
- Position
- School
- Height
- Weight
- 40-yard Dash
- Vertical Jump
- Bench Press
- Broad Jump
- 3-Cone Drill
- 20-yard Shuttle

Pro Football Reference

- Team
- Round
- Pick
- Year
- Weighted AV (response variable)

OLS Regression

Reason:

- Simple and easily interpretable

Key Results:

- RMSE = 18.14819
- R^2 = 0.2943186

Insights:

- Position and cone drill were the most significant predictors
- Vertical jump, height, weight, 20-yard shuttle, year, and bench press were insignificant predictors

Coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	134.089245	18.686476	7.176	1.03e-12	***
PositionCB	-23.259505	3.887341	-5.983	2.60e-09	***
PositionDB	8.971026	18.495344	0.485	0.627702	
PositionDE	-12.450604	3.540882	-3.516	0.000448	***
PositionDL	-15.670014	4.013128	-3.905	9.76e-05	***
PositionDT	-5.226052	3.333109	-1.568	0.117066	
PositionEDGE	-20.572824	4.809590	-4.277	1.98e-05	***
PositionFB	-19.962325	5.609199	-3.559	0.000382	***
PositionILB	-16.442595	3.895125	-4.221	2.54e-05	***
PositionK	-9.578149	6.870133	-1.394	0.163428	
PositionLB	-16.833377	4.413277	-3.814	0.000141	***
PositionLS	-15.764114	10.889811	-1.448	0.147893	
PositionOG	-4.437432	3.498400	-1.268	0.204804	
PositionOL	-10.141681	3.971249	-2.554	0.010733	*
PositionOLB	-13.456433	3.744549	-3.594	0.000334	***
PositionOT	-3.554094	3.286164	-1.082	0.279597	
PositionP	-13.851171	5.339813	-2.594	0.009561	**
PositionQB	-9.961936	3.709188	-2.686	0.007300	**
PositionRB	-18.222999	3.864227	-4.716	2.58e-06	***
PositionS	-18.693887	3.843773	-4.863	1.25e-06	***
PositionTE	-21.012583	3.768159	-5.576	2.81e-08	***
PositionWR	-20.878908	3.862097	-5.406	7.25e-08	***
X40.yd.Dash	-9.517519	3.217648	-2.958	0.003135	**
X3.Cone.Drill	-5.246275	1.865512	-2.812	0.004970	**
Pick	-0.151819	0.006134	-24.749	< 2e-16	***

LASSO

Reason:

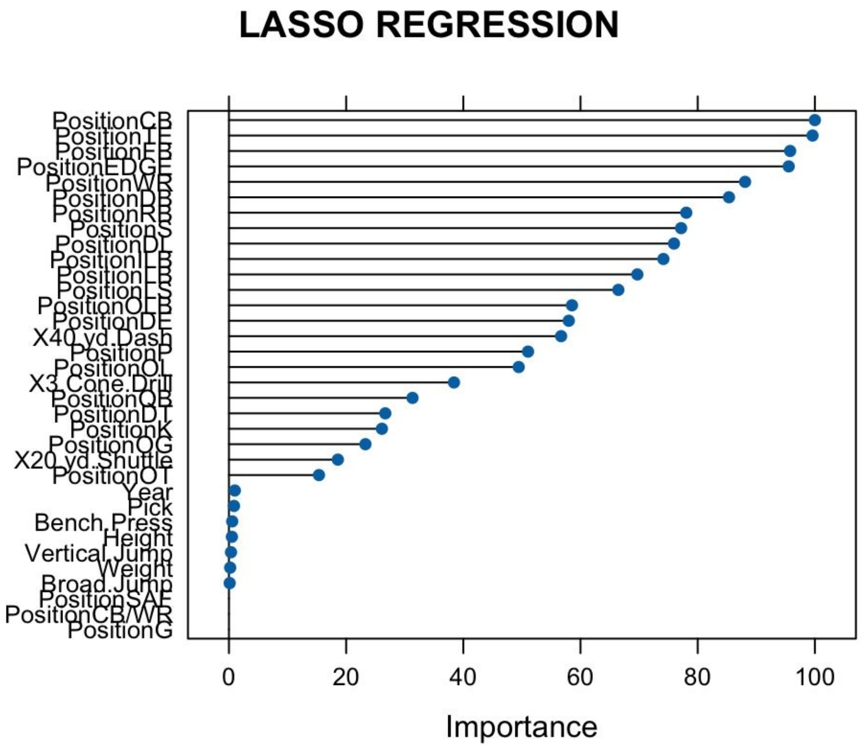
- Identifies the most important predictors

Key Results:

- RMSE = 19.88293
- R^2 = 0.241662

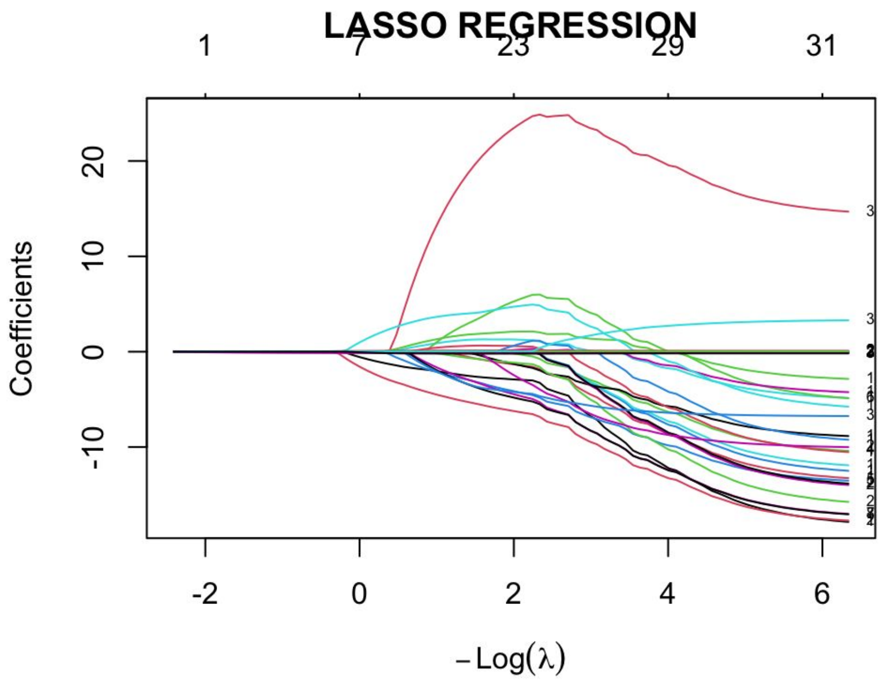
Insights:

- LASSO identified position, 40 yard dash, and 3-cone drill as the most important predictors



LASSO

- As the λ decreases, more predictors enter the model, but many coefficients shrink toward zero, showing that most variables have limited independent predictive power.
- Only a small subset of features retain non-zero coefficients across λ values, indicating that a few variables drive most of the predictive signal.
- The rapid collapse of many coefficient paths to zero suggests strong redundancy and correlation among combine metrics, meaning lasso effectively performs feature selection by removing weak or overlapping predictors.



Ridge

Reason:

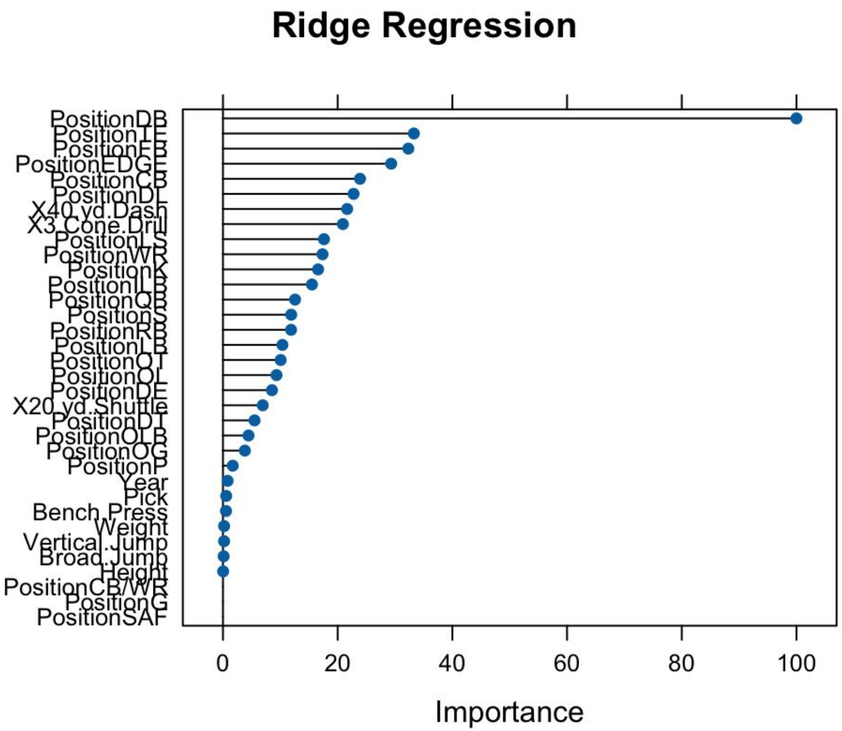
- Handles multicolleanarity and stablizes prediction

Key Results:

- RMSE = 19.86566
- R^2 = 0.2409259
- Best lambda = 101.072267

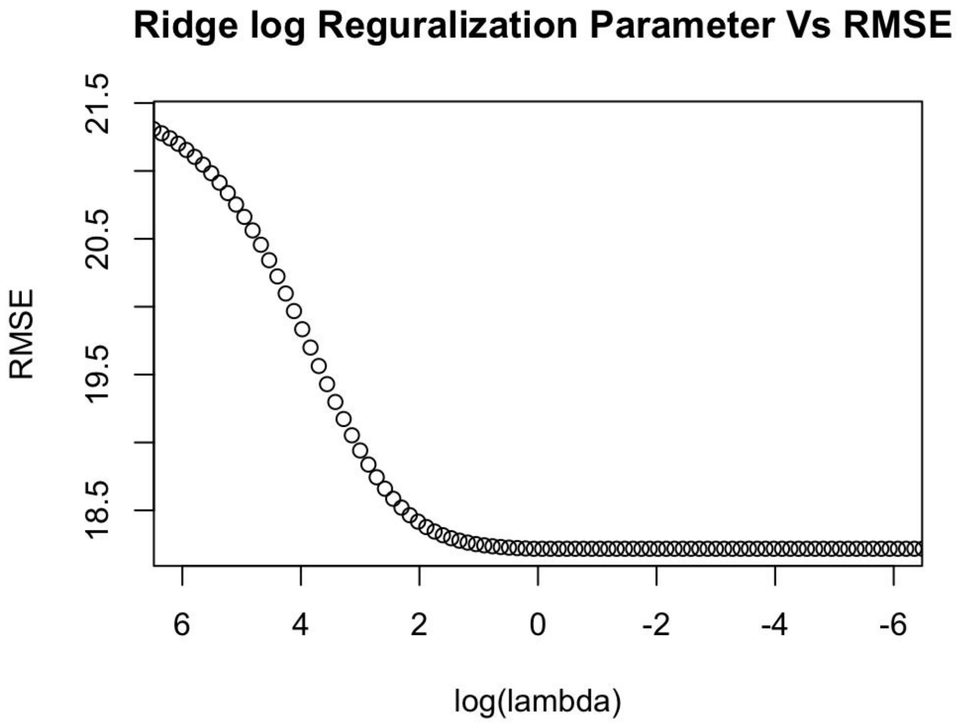
Insights:

- Ridge improved prediction accuracy slightly over OLS by reducing variance
- Position, 40 yard dash, and 3-cone drill are the most significant predictors



Ridge

- Strong regularization leads to worse performance, with RMSE starting around 21.5, indicating underfitting when coefficients are overly shrunk.
- As λ decreases, RMSE drops sharply, showing that reducing regularization allows the model to better capture signal in the data and improve predictive accuracy.
- Performance stabilizes around RMSE ≈ 18.3 at lower λ values, suggesting diminishing returns beyond a certain point and an optimal ridge penalty region where bias–variance tradeoff is balanced.



Random Forest

Reason:

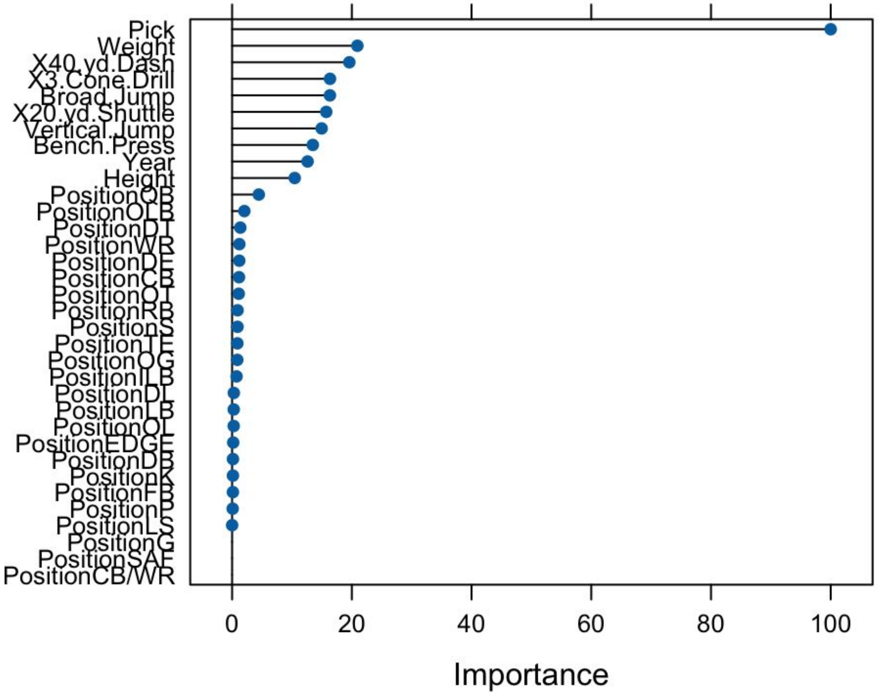
- Handles complex patterns and no linear assumptions

Key Results:

- RMSE = 17.62961
- R^2 = 0.3335632
- Best mtry = 10

Insights:

- Pick, weight, and 40 yard dash are the most important predictors



Best Model

Model	RMSE	R^2
OLS	18.14819	0.2943186
Lasso	19.88293	0.241662
Ridge	19.86566	0.2409259
Random Forest	17.62961	0.3335632

- Through our results, we can see the random forest model was the best

Limitations

- Raw prediction model, does not account for team/scheme fit
- Data availability; not every player participates in every combine event
- Undrafted players are not accounted for, even though many undrafted players have had very high AV's

Conclusion

- Random forest produced the best model
- 40-yard dash and 3-cone drill were significant predictors in all the models
- Bench press, broad jump, and vertical jump may not be as significant as predictors as one would think

Recommendation

The models can be useful starting points or stepping stones for NFL scouts when evaluating players. The model also can be used for younger players who are only a few years removed from the combine, projecting where their careers will/can go, which can be useful in contract and trade negotiations. Overall, the models will be most productive when used as supplementary tools.